OFFICIAL DOCUMENTATION · DATAUM

DataUA User Guidebook

A complete step-by-step guide to uploading data, configuring assessments, generating reports, and exporting results.

DATAUM

Beta Vo.37

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INTRODUCTION

What is DataUA?

DataUA (Data University Assistant) is a **self-contained, browser-based application** that transforms raw Excel or CSV grade sheets into formatted, printable assessment reports — with no installation required and no data ever leaving your device.

It is designed for instructors, TAs, and coordinators who need to process student assessment data quickly, apply a grading formula, visualize performance, and export the results — and is developed by **Dataum**, a startup focused on building customized, feature-rich applications for universities, factories, and companies.

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PRIVACY FIRST

All data is processed entirely in your browser. Nothing is sent to any server. You can even use this application offline once the page has loaded.

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WHAT YOU NEED BEFORE STARTING

An Excel (.xlsx or .xls) or CSV file containing your student data. At minimum you need a column with student names and at least one column with numeric scores or grades.

01

STEP ONE

Upload Your Data File

UPLOAD

1

Uploading an Excel or CSV File

On the first screen you will see a large upload area with a file picker. Click it to browse your computer, or simply drag and drop your file onto it.

- Supported formats: **.xlsx**, **.xls**, and **.csv**
- After selecting a file, the application will read it immediately — no "submit" button needed
- Once the file is parsed, you are automatically taken to Step 2

FILE FORMAT TIPS

If your Excel file has multiple sheets, only the first sheet is read. Make sure your data is on the first sheet before uploading.

DON'T HAVE A FILE YET?

You can export any spreadsheet from Google Sheets as .xlsx or .csv via File → Download. For Moodle grade books, use the "Export → Excel spreadsheet" option.

02

STEP TWO

Select the Header Row

HEADERS

2

Click the Row That Contains Your Column Names

The tool displays all rows from your file in a table. Your job here is to tell it which row holds the column headers (e.g., "Name", "Score", "Section").

- Click anywhere on a row to highlight and select it as the header row
- The selected row will be highlighted in blue
- Everything above the selected row is ignored (useful if your file has a title or logo rows at the top)
- Once selected, the app automatically advances to Step 3

WHY THIS STEP EXISTS

Many real-world spreadsheets have a title, institution name, or date in the first few rows before the actual data. This step lets you skip those rows and pick the correct header row no matter where it appears.

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STEP THREE

Edit Your Data

EDIT

3

Fix Errors Before Processing

The tool now shows your data in an editable table. You can correct typos, missing values, or wrong scores before the report is generated.

- **Click any cell** to open a small editing dialog for that cell
- Type the new value and click *Save*
- You can also click column headers to sort the data
- When you are satisfied, click the green **"Done Editing"** button at the top right
- Clicking *Done Editing* advances you to Step 4 automatically

IT'S OKAY TO SKIP EDITING

If your data is already clean and correct, just click "Done Editing" immediately to proceed. This step only exists for convenience — it's not mandatory to change anything.

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STEP FOUR

Choose Display Columns

COLUMNS

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Select What Appears in the Final Results Table

Your spreadsheet might have many columns (Student ID, Phone, Section, etc.). Here you decide which ones should appear in the final report — only the columns you care about.

- The left panel lists all available columns from your file
- Tick the checkbox next to each column you want to include
- Selected columns appear in the right panel as draggable cards
- **Drag to reorder** the right panel cards — the order here becomes the column order in the report
- Use the search box to quickly find columns if you have many
- Click **"Select All"** to include every column at once
- Once at least one column is selected, the **"Continue to Configuration"** button activates — click it to proceed

THE CALCULATED SCORE IS ALWAYS ADDED AUTOMATICALLY

You don't need to select the score column here. The tool will always append a "Score" and "Status" column at the end of your chosen display columns after calculation.

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STEP FIVE

Configure Column Roles & Grading Mode

This is the most important step. You tell the app two things: (1) what each column in your data *means*, and (2) how scores should be calculated.

⚙ CONFIGURE

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Part A — Column Mapping (Assign Roles to Columns)

The right side of the screen shows your data table. **Click any column header** to open a popup where you assign that column a role:

- **Name** — The column containing student names (required)
- **Add (+)** — A numeric column to be added to the score (e.g., quiz marks, attendance points)
- **Subtract (-)** — A numeric column to be subtracted (e.g., penalty deductions)
- **Ignore** — A column that exists in the data but should not affect the score at all
- You can also set a **Weight** (multiplier) and an **Offset** (fixed bonus/penalty) per column

The left panel (*Configuration Panel*) shows a live summary of all assigned column roles. The **"Generate Report"** button activates once at least one Name column and one Add column have been assigned.

📊 GRADING

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Part B — Grading Mode

Below the column mapping section in the left panel, choose a grading mode. Each mode changes how the final score is interpreted and what badges/labels appear on students.

After configuring both parts, click the large **"GENERATE REPORT"** button. The app will calculate all scores and take you to Step 6 — Results.

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REQUIRED: AT LEAST ONE "NAME" AND ONE "ADD" COLUMN

The "Generate Report" button stays disabled until you assign the Name role to one column and the Add role to at least one numeric column.

06

STEP SIX

Results, Charts & Export

📊 RESULTS

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Reading the Results Screen

The results page shows four summary statistics at the top: Total Students, Average Score, Highest Score, and Lowest Score. Below that you'll find:

- **Performance Chart** — A visual chart of scores. Use the tabs to switch between chart types (Bar, Pie, Line, Radar, Doughnut, Horizontal Bar, Polar Area, Bubble, Scatter, Area, Mixed).
- **Multi-Chart Mode** — Toggle this on to display up to 8 chart types simultaneously in a 2-column grid.
- **Table Display Options** — Switch between 8 different views of your data using the tabs above the results table (see below for details).
- **Analytics Panel** — Deeper statistical breakdowns below the table: grade distribution, pass/fail ratios, score bands, and percentile analysis.

TABLE VIEWS

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The 8 Table View Modes

The **Table Display Options** tabs let you switch between eight different ways to view and analyse your results. You can switch freely at any time — the view you have active is exactly what gets exported to PDF and Print.

VIEW	WHAT IT SHOWS
Ranked List	All students sorted by score, with rank numbers and medal icons for the top 3. Includes all display columns selected in Step 4.
Categorized	Students grouped into cards by their status category (e.g. Pass, Fail, Excellent). Each card shows selected column values.
Detailed View	Expanded individual card per student showing all selected column data in a structured layout.
Trend Analysis	Card-based view showing each student's performance trend (Improving / Stable / Needs Support), score, status, and rank with a progress bar.
Scatter Plot	Visual dot plot where each student is a circle sized and coloured by their score. Useful for spotting distribution and outliers at a glance.
Executive Summary	One-page overview with key headline numbers (Excellent count, Pass count, Class Average, At-Risk count), written insights, and top 5 performers.
Comparison Matrix	Full table comparing all students with score, rank, status badge, a visual progress bar, and deviation from the class average.
Statistical Dashboard	Descriptive statistics (mean, median, mode, standard deviation, variance, range) plus a score distribution histogram.

💾 EXPORT

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Exporting & Saving Your Report

The toolbar at the top of the Results page has five action buttons:

BUTTON	WHAT IT DOES	OUTPUT
💾 **Save**	Saves results to browser memory (localStorage) for later retrieval	Stored in browser — no file created
CSV	Exports the results table as a comma-separated values file	.csv file download
Excel	Exports the results table as a formatted Excel workbook	.xlsx file download
PDF	Generates a full report: header, statistics, chart(s), and the active table view — all faithfully captured as they appear on screen	.pdf file download
Print	Opens a print-ready version in a new window with the same faithful layout capture as PDF — all 8 table views print correctly	Browser print dialog

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WHAT GETS EXPORTED

Both PDF and Print capture whichever **TABLE VIEW** is currently active — including complex visual views like the Scatter Plot, Executive Summary, Statistical Dashboard, and Trend Analysis. Switch to the view you want before clicking Export or Print.

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LOADING PREVIOUSLY SAVED RESULTS

At any time, click the **"LOAD SAVED"**

button in the top navigation bar to restore a previously saved assessment without re-uploading any files.

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STARTING A NEW ASSESSMENT

To process a completely different file, click **"NEW ASSESSMENT"** in the top navigation bar. This reloads the page and clears all current data.

07

REFERENCE

All Grading Modes Explained

The tool offers 17 grading modes. Choose the one that matches your assessment context.

Pass / Fail

Students are labelled Pass or Fail based on a configurable minimum score threshold.

Behavioral

Evaluates scores on behavioral/soft-skills criteria with descriptive labels.

Weighted

Each score column can have a different weight. The final score is a weighted average.

Creative

Designed for creative projects; applies bonus multipliers for exceptional scores.

Project

Treats columns as project milestones; milestone weights are configurable.

Percentile

Ranks students by percentile position within the class rather than raw score.

Letter Grade

Converts numeric scores to standard letter grades (A, B, C, D, F).

Custom

Define your own score bands and labels for maximum flexibility.

GPA

Converts scores to GPA points on a 4.0 scale using standard mapping.

Rubric

Multi-criteria rubric scoring; each column is a rubric dimension.

Z-Score

Standardises scores relative to the class mean and standard deviation.

Contest / Competition

Designed for competitions; students are awarded ranks and medals (Gold/Silver/Bronze).

Star Rating

Converts scores to a 1–5 star visual rating system.

Mastery

Curve

Categorises students as Mastered, Developing, or Beginning based on thresholds.

Applies a statistical bell-curve adjustment to normalize class scores.

Olympic

Ranks students with Olympic-style scoring; ties handled by time/secondary criteria.

Portfolio

Four weighted dimensions: Technical (default 40%), Creativity (30%), Presentation (20%), Documentation (10%).

Gamified

Assigns XP (experience points) and achievement levels in a game-style format.

PORTFOLIO MODE TIP

In Portfolio mode, a "Validate Weights" button appears. Make sure Technical + Creativity + Presentation + Documentation adds up to exactly 100% before generating the report.

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TIPS & TRICKS

Keyboard Shortcuts & Power User Features

`Ctrl + S`	Save current results to browser storage
`Ctrl + P`	Print / open print dialog (results only)
`Ctrl + E`	Export results to CSV

Theme & Appearance

The app includes four built-in themes: **Light** (default), **Dark**, **DataUA Official** (branded navy/orange theme), and **High Contrast** (accessibility). Click the floating palette button at the bottom-right of the screen to switch themes.

The **right-side customizer panel** (tab on the right edge of the screen) gives you full granular control split across three tabs:

- **Colors tab** — Customize every color group individually: Background Colors, Accent Colors, Text Colors, Status Colors, Border color, and **Input Boxes** (background, text color, and border of all input fields — including the Edit Cell box).
- **Layout tab** — Choose interface density (Comfortable / Cozy / Compact) and toggle split-view results layout on wide screens.
- **Comfort tab** — Toggle reduced motion for users who prefer minimal animations.

All customizations are saved automatically to browser storage and restored on next visit. Use the **Reset** button to return to defaults.

Save & Load Sessions

After generating results, use the **Save** button to store your assessment in the browser. Your saved assessments persist even after closing the tab (until browser data is cleared).

To reload a saved session without re-uploading your file, click **"Load Saved"** from the top navigation bar. A modal will appear listing all saved assessments with timestamps. You can load or delete each one.

Multi-Chart View

On the Results page, look for the **"Multi-Chart Mode"** toggle. When enabled, a panel appears letting you tick which chart types to display simultaneously. Up to 8 chart types can be shown in a 2-column grid layout.

Table View Modes

The results table supports **8 different view modes**, switchable via the tabs above the table. Each view presents your data differently and all 8 are fully exportable via PDF and Print:

Ranked List · Categorized · Detailed View · Trend Analysis · Scatter Plot · Executive Summary · Comparison Matrix · Statistical Dashboard.

See **Section 06** → **Table Views** for a full description of each mode.

Analytics Tab

The **Analytics** panel appears below the results table automatically after generation. It shows deeper breakdowns: grade distribution histograms, pass/fail ratios, score range analysis, and a percentile table.

DataUA — Data University Assistant

Designed and developed by **Dataum** — building customized, widely-featured applications for universities, factories, and companies. All data processed locally. No internet connection required.